

Integration by Parts, Chosen Strategically

Single and repeated integration by parts, parts for logarithms and inverse functions, and parts inside definite and improper integrals.

- Parts is $\int u dv = uv - \int v du$. It earns its keep only when $\int v du$ is *easier* than what you started with, so state your u and dv before you compute anything.
- LIATE is a tie-breaker, not a law: choose u from the first category present — Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — and dv from the last.
- Antiderivatives need $+C$; definite integrals do not.
- For an improper integral, replace the infinite endpoint by a letter, integrate, and take the limit. Write the limit; do not substitute infinity directly.
- Leave exact answers exact: π , e , and fractions, not decimals.

1. Evaluate $\int x e^x dx$. State your choice of u and dv before integrating.

Answer: _____

2. Evaluate $\int x \cos x \, dx$.

Answer: _____

3. Evaluate $\int \ln x \, dx$.

Strategy hint: there is only one factor, so the choice of dv is forced.

Answer: _____

4. Evaluate $\int x^2 e^x dx$. One application of parts will not finish it — say what the remaining integral is, then handle it.

Answer: _____

5. Evaluate $\int_0^1 x e^{2x} dx$. Give the exact value.

Answer: _____

6. Evaluate $\int x \ln x \, dx$. Explain in one line why $u = \ln x$ rather than $u = x$.

Answer: _____

7. Evaluate $\int x \sec^2 x \, dx$. Choose the factor whose antiderivative you already know as dv .

Answer: _____

8. Evaluate $\int e^x \sin x \, dx$. Parts twice returns you to the integral you started with — treat that integral as an unknown and solve for it.

Answer: _____

9. Evaluate $\int_0^\pi x^2 \cos x \, dx$. Give the exact value.

Answer: _____

10. Evaluate $\int_1^\infty \frac{\ln x}{x^2} \, dx$, or show that it diverges. Replace the infinite limit by a finite t , integrate by parts, and then let $t \rightarrow \infty$.

- (a) Parts on $\int_1^t \frac{\ln x}{x^2} dx$ produces the boundary term $-\frac{\ln t + 1}{t}$. Find $\lim_{t \rightarrow \infty} \left(-\frac{\ln t + 1}{t}\right)$, and say which growth rate wins.

Answer: _____

- (b) Use part (a) to state the exact value of the improper integral.

Answer: _____